# BDNet:Bio-Inspired Dual-Backbone Small Object Detection Network

## Supplementary Material

## A. Construction of ESCK

Cross-orientation suppression (COS) [1, 2] is a key neural modulation mechanism in the visual cortex. Its core principle is as follows: when a neuron prefers a specific orientation (such as vertical), if stimuli of other orientations (such as horizontal) appear in its receptive field, inhibitory interneurons reduce the firing rate of that neuron. This enhances the response to the target orientation signal while suppressing interference from non-preferred orientations. This mechanism has been extensively studied and confirmed in the primary visual cortex (V1 area) and is one of the fundamental bases for achieving orientation selectivity and information filtering in V1 neurons.

To simulate the COS mechanism, we designed eight ESCKs (Enhanced Selective Convolutional Kernels) in different directions. Inspired by the concept of pixel difference convolution [3], ESCK transforms the pixel difference in the output into a difference in convolutional kernel weights. This enables the convolution operation to achieve a response pattern that enhances same-orientation inputs and suppresses cross-orientation inputs.

Suppose the sliding operation of the original $3 \times 3$ convolution kernel in the spatial dimension can be described as:

$$\begin{aligned} y &= \sum_{i=1}^{9} w_i \cdot x_i = W \cdot X \\ &= \begin{bmatrix} w_1 & w_2 & w_3 \\ w_4 & w_5 & w_6 \\ w_7 & w_8 & w_9 \end{bmatrix} \cdot \begin{bmatrix} x_1 & x_2 & x_3 \\ x_4 & x_5 & x_6 \\ x_7 & x_8 & x_9 \end{bmatrix} \end{aligned} \tag{1}$$

0) To simulate the response characteristics of COS in the $180^\circ$ direction, i.e., enhancing co-oriented stimuli and suppressing cross-oriented stimuli, the pixel output is defined as:

$$y' = W \cdot X' \tag{2}$$

$$X' = \begin{bmatrix} x_1 - x_4 & x_2 - x_5 & 0 \\ x_4 + x_1 + x_7 & x_5 + x_2 + x_8 & 0 \\ x_7 - x_4 & x_8 - x_5 & 0 \end{bmatrix} \tag{3}$$

Based on this, the differential computation form of the convolution kernel can be derived inversely as:

$$y' = W' \cdot X \tag{4}$$

$$W' = \begin{bmatrix} w_1 + w_4 & w_2 + w_5 & 0 \\ w_4 - w_1 - w_7 & w_5 - w_2 - w_8 & 0 \\ w_7 + w_4 & w_8 + w_5 & 0 \end{bmatrix} \tag{5}$$

Similarly, we sequentially derived the pixel outputs and convolutional kernel difference forms for the $135^\circ$, $90^\circ$, $45^\circ$, $0^\circ$, $-45^\circ$, $-90^\circ$ and $-135^\circ$ directions. The specific process is as follows:

1) In the $135^\circ$ direction, the $X'$ is:

$$\begin{bmatrix} x_1 + \frac{x_2}{2} + \frac{x_4}{2} & x_2 - \frac{x_1}{2} - \frac{x_5}{2} & x_3 - x_5 \\ x_4 - \frac{x_1}{2} - \frac{x_5}{2} & x_5 + \frac{x_2}{2} + \frac{x_4}{2} + x_3 + x_7 & 0 \\ x_7 - x_5 & 0 & 0 \end{bmatrix} \tag{6}$$

The $W'$ is:

$$\begin{bmatrix} w_1 - \frac{w_2}{2} - \frac{w_4}{2} & w_2 + \frac{w_1}{2} + \frac{w_5}{2} & w_3 + w_5 \\ w_4 + \frac{w_1}{2} + \frac{w_5}{2} & w_5 - \frac{w_2}{2} - \frac{w_4}{2} - w_3 - w_7 & 0 \\ w_7 + w_5 & 0 & 0 \end{bmatrix} \tag{7}$$

2) In the $90^\circ$ direction, the $X'$ is:

$$\begin{bmatrix} x_1 - x_2 & x_2 + x_1 + x_3 & x_3 - x_2 \\ x_4 - x_5 & x_5 + x_4 + x_6 & x_6 - x_5 \\ 0 & 0 & 0 \end{bmatrix} \tag{8}$$

The $W'$ is:

$$\begin{bmatrix} w_1 + w_2 & w_2 - w_1 - w_3 & w_3 + w_2 \\ w_4 + w_5 & w_5 - w_4 - w_6 & w_6 + w_5 \\ 0 & 0 & 0 \end{bmatrix} \tag{9}$$

3) In the $45^\circ$ direction, the $X'$ is:

$$\begin{bmatrix} x_1 - x_5 & x_2 - \frac{x_3}{2} - \frac{x_5}{2} & x_3 + \frac{x_2}{2} + \frac{x_6}{2} \\ 0 & x_5 + x_1 + x_9 + \frac{x_2}{2} + \frac{x_6}{2} & x_6 - \frac{x_3}{2} - \frac{x_5}{2} \\ 0 & 0 & x_9 - x_5 \end{bmatrix} \tag{10}$$

The $W'$ is:

$$\begin{bmatrix} w_1 + w_5 & w_2 + \frac{w_3}{2} + \frac{w_5}{2} & w_3 - \frac{w_2}{2} - \frac{w_6}{2} \\ 0 & w_5 - w_1 - w_9 - \frac{w_2}{2} - \frac{w_6}{2} & w_6 + \frac{w_3}{2} + \frac{w_5}{2} \\ 0 & 0 & w_9 + w_5 \end{bmatrix} \tag{11}$$

4) In the $0^\circ$ direction, the $X'$ is:

$$\begin{bmatrix} 0 & x_2 - x_5 & x_3 - x_6 \\ 0 & x_5 + x_2 + x_8 & x_6 + x_3 + x_9 \\ 0 & x_8 - x_5 & x_9 - x_6 \end{bmatrix} \tag{12}$$

The $W'$ is:

$$\begin{bmatrix} 0 & w_2 + w_5 & w_3 + w_6 \\ 0 & w_5 - w_2 - w_8 & w_6 - w_3 - w_9 \\ 0 & w_8 + w_5 & w_9 + w_6 \end{bmatrix} \tag{13}$$

5) In the $-45^\circ$ direction, the $X'$ is:

$$\begin{bmatrix} 0 & 0 & x_3 - x_5 \\ 0 & x_5 + x_3 + x_7 + \frac{x_6}{2} + \frac{x_8}{2} & x_6 - \frac{x_5}{2} - \frac{x_9}{2} \\ x_7 - x_5 & x_8 - \frac{x_5}{2} - \frac{x_9}{2} & x_9 + \frac{x_6}{2} + \frac{x_8}{2} \end{bmatrix} \tag{14}$$

The $W'$ is:

$$\begin{bmatrix} 0 & 0 & w_3 + w_5 \\ 0 & w_5 - w_3 - w_7 - \frac{w_6}{2} - \frac{w_8}{2} & w_6 + \frac{w_5}{2} + \frac{w_9}{2} \\ w_7 + w_5 & w_8 + \frac{w_5}{2} + \frac{w_9}{2} & w_9 - \frac{w_6}{2} - \frac{w_8}{2} \end{bmatrix} \tag{15}$$

6) In the $-90^\circ$ direction, the $X'$ is:

$$\begin{bmatrix} 0 & 0 & 0 \\ x_4 - x_5 & x_5 + x_4 + x_6 & x_6 - x_5 \\ x_7 - x_8 & x_8 + x_7 + x_9 & x_9 - x_8 \end{bmatrix} \tag{16}$$

The $W'$ is:

$$\begin{bmatrix} 0 & 0 & 0 \\ w_4 + w_5 & w_5 - w_4 - w_6 & w_6 + w_5 \\ w_7 + w_8 & w_8 - w_7 - w_9 & w_9 + w_8 \end{bmatrix} \tag{17}$$

7) In the $-135^\circ$ direction, the $X'$ is:

$$\begin{bmatrix} x_1 - x_5 & 0 & 0 \\ x_4 - \frac{x_5}{2} - \frac{x_7}{2} & x_5 + x_1 + x_9 + \frac{x_4}{2} + \frac{x_8}{2} & 0 \\ x_7 + \frac{x_4}{2} + \frac{x_8}{2} & x_8 - \frac{x_5}{2} - \frac{x_7}{2} & x_9 - x_5 \end{bmatrix} \tag{18}$$

The $W'$ is:

$$\begin{bmatrix} w_1 + w_5 & 0 & 0 \\ w_4 + \frac{w_5}{2} + \frac{w_7}{2} & w_5 - w_1 - w_9 - \frac{w_4}{2} - \frac{w_8}{2} & 0 \\ w_7 - \frac{w_4}{2} - \frac{w_8}{2} & w_8 + \frac{w_5}{2} + \frac{w_7}{2} & w_9 + w_5 \end{bmatrix} \tag{19}$$

## B. Experimental Support

**Experimental Setup.** The baseline model for small object detection constructed in this study is based on the Ultralytics framework (version 8.3.78), with the optimized YOLO12 architecture as its core, as shown in Tab. A. It is implemented in a PyTorch 2.7.1 environment. The hardware configuration consists of a single NVIDIA RTX 3090Ti GPU with 24GB VRAM (CUDA 12.4 + cuDNN 9.5.1). For data augmentation, Mosaic augmentation, HSV jitter (hsv_h: 0.015; hsv_s: 0.7; hsv_v: 0.4), horizontal flipping (p=0.5), and random erasing (p=0.4) were employed to suppress overfitting.

Regarding training parameters, input images were uniformly resized to 640×640, with a batch size of 8. The optimizer used was SGD, with an initial learning rate of 0.01 and a warm-up phase lasting three epochs. After warm-up, the learning rate was adjusted according to the scheduler, decreasing by $2 \times 10^{-5}$ per epoch. The total number of training epochs was 300. Our model did not use any pretrained weights during the entire training process.

Table A. Optimization details.

| Component | YOLO12 | Optimization details |
|---|---|---|
| Backbone | From P1 to P5 | • From P1 to P4<br>• P5 (Conv and A2C2f) removed |
| Neck | P4-P3-P4-P5 | • P3-P2-P3<br>• Downsampling (C3k2) removed<br>• Conv channels: 512<br>• A2C2f channels: [512, 1024, 1024] |
| Head | P3, P4, P5 | P2, P3 |

**Datasets.** The VisDrone2019, NWPU VHR-10, and AI-TODv2 datasets selected for this study exhibit significant differences in terms of object size, background complexity, and shooting perspective, enabling a comprehensive evaluation of the model's adaptability and robustness across various remote sensing scenarios.

The VisDrone2019 dataset originates from remote sensing images captured from a drone perspective, simulating real-world drone inspection scenarios. It is characterized by typical small objects, dense distributions, and multiple occlusions. Most object sizes are smaller than $32 \times 32$ pixels, and they are noticeably affected by illumination variations and perspective distortion. The dataset covers 10 object categories, including pedestrians and vehicles. According to the official standard, it is divided into 6,471 training images, 548 validation images, and 1,610 test images, totaling approximately 2.6 million annotated objects. The diversity and complexity of this dataset make it an important benchmark for evaluating model performance in complex urban scenarios [4].

The NWPU VHR-10 dataset consists of very high-resolution (VHR) satellite imagery. Its main features are the large contrast differences between objects and the background, as well as color or shape confusion in some categories (e.g., storage tanks, tennis courts), thereby testing the model's recognition capability under high-interference backgrounds. This dataset contains 10 categories of artificial objects, such as airplanes and ships, with a total of 650 positive sample images containing object annotations and 150 negative sample images without targets. For the positive sample portion, the commonly used academic split ratio of 8:1:1 was adopted, resulting in 520 training images, 65 validation images, and 65 test images [5].

The AI-TODv2 dataset focuses on aerial tiny object detection. Its core characteristic is that the average object size is only 12.8 pixels, significantly smaller than other remote sensing datasets, and it contains a large number of sub-pixel objects, making it highly challenging. This dataset covers 8 object categories, including airplanes and bridges, with a uniform image resolution of $800 \times 800$ pixels. According

to the official split, the trainval set contains 11,214 training images and 2,804 validation images, while the test set contains 14,018 test images, totaling 700,621 annotated instances. This dataset provides an ideal testbed for evaluating model detection capability in very low-resolution and extremely small object scenarios [6].

**Evaluation Metrics.** To comprehensively quantify model performance, this study adopts general evaluation metrics from the object detection field [7]. Following previous work, the NWPU VHR-10 dataset primarily uses $mAP_{50}$ (%) (mean Average Precision at an IoU threshold of 0.5) as the core metric to highlight the model's detection accuracy for small objects and in high-resolution scenes. For the VisDrone2019 dataset, based on $mAP_{50}$ (%), the comprehensive metric $mAP_{50-95}$ (%) (mean Average Precision over IoU thresholds from 0.5 to 0.95) is further introduced to more comprehensively evaluate the model's detection performance in complex urban scenarios. Additionally, the number of parameters (Params, M) and computational complexity (GFLOPs) are reported to evaluate the model's inference efficiency and lightweight characteristics. The evaluation for the AI-TODv2 dataset follows the COCO official evaluation standard [8] to calculate AP (%) (overall Average Precision), $AP_{50}$ (%) (AP at IoU=0.5), and $AP_{75}$ (%) (AP at IoU=0.75). Furthermore, to analyze the model's performance across different object scales, based on the dataset's scale division criteria, $AP_{vt}$ (%) (Average Precision for objects $\leq$ 8 pixels), $AP_t$ (%) (8–16 pixels), $AP_s$ (%) (16–32 pixels), and $AP_m$ (%) (32–64 pixels) are further calculated to comprehensively reveal the model's detection performance across tiny, very small, small, and medium-sized objects.

## C. Supplemental Analysis and Verification

This section provides visual analysis of feature maps related to the core contributions of the main text. It also explores the diversity of color spaces, validates the extension to detection tasks, evaluates key component performance, and assesses generalization capabilities and deployment value, serving as a robust supplement to the experimental analysis in the main text.

**Visual Analysis of Feature Maps.** To clearly demonstrate the model's enhancement of key target features, we extracted and visualized feature maps before and after processing by CAM, VCHM, OrSM, and FFM, as shown in Fig. A. CAM and VCHM significantly improve color contrast, which manifests as strong activations in high-contrast regions on the feature maps, while values in low-contrast regions change smoothly, exhibiting consistent responses across the area. Furthermore, to demonstrate that OrSM possesses stronger orientation selectivity, we conducted a frequency domain analysis. As shown in Fig. B, in the frequency domain, line brightness reflects response intensity for specific orientations, confirming that OrSM exhibits powerful orientation selectivity.

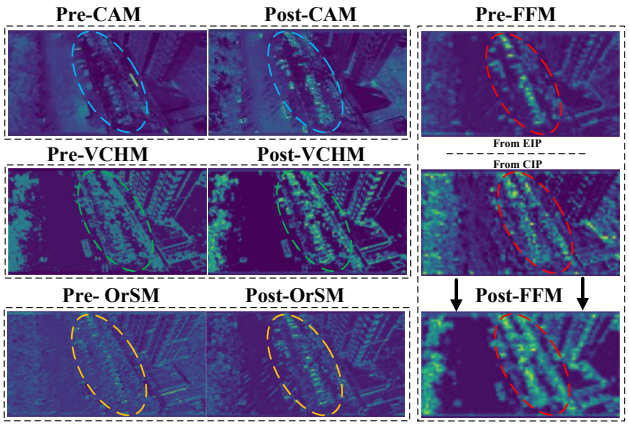



Figure A. Feature Map Visualization: Pre- and Post-Processing by CAM, VCHM, OrSM, and FFM.

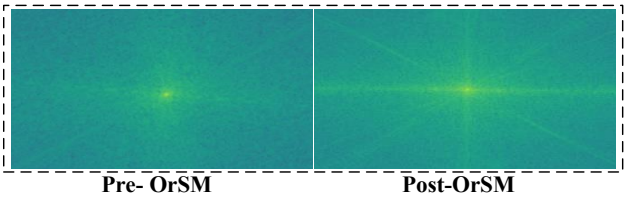



Figure B. Frequency Domain Analysis.

**Exploration of Color Space Diversity.** Regarding the exploration of color space diversity, we conducted experiments across different color spaces, including RGB, HSV, Lab, and Opponent (RG/BY). As shown in Tab. B, the results demonstrate that our method performs best when processing RGB information directly.

Table B. Color Space Diversity Exploration on the VisDrone2019 Dataset.

| Method | $mAP_{50}$ | Params | GFLOPs |
|---|---|---|---|
| HSV | 48.7 | 2.59 | 52.44 |
| Lab | 49.1 | 2.59 | 52.44 |
| Opponent (RG/BY) | 50.2 | 2.59 | 52.44 |
| RGB | 50.5 | 2.59 | 52.44 |

**Validation of Extension to Detection Tasks.** As shown in Tab. C, we compared the detection performance of horizontal bounding boxes (HBB) and oriented bounding boxes (OBB), validating the model's potential for adaptation to rotated objects.

**Performance Evaluation of Key Components.** While keeping the parameter count consistent, we conducted equivalent replacement experiments using standard convolutions to demonstrate that the color opponency of CAM,

Table C. Performance comparison of different detection heads with the same baseline model on HBB (VisDrone2019) and OBB (DOTA-v1.5) datasets.

| Method | $mAP_{50}$ | Params | GFLOPs |
|---|---|---|---|
| HBB | 47.8 | 1.98 | 45.98 |
| HBB (Add OrSM) | 48.3 (↑0.5) | 2.04 | 47.13 |
| OBB | 43.9 | 2.35 | 57.93 |
| OBB (Add OrSM) | 44.8 (↑0.9) | 2.41 | 59.08 |

the hue enhancement of VCHM, and the orientation selectivity of OrSM directly contribute to the accuracy improvement, as shown in Tab. D. Furthermore, through comparative experiments with mainstream strategies such as Concat Plus 1×1 Conv, Cross-attention, and SE-style cross gating, we validated the superiority of the proposed FFM strategy in terms of both accuracy and efficiency, as presented in Tab. E. For multi-scale fusion methods, we conducted ablation experiments as shown in Tab. F, and the results demonstrate that our multi-scale fusion method is both simple and efficient.

Table D. Replacement Ablation Experiments.

| Method | $mAP_{50}$ | Params | GFLOPs |
|---|---|---|---|
| CAM | 48.0 | 1.98 | 46.39 |
| CAM-Replace | 47.3 (↓0.7) | 1.98 | 46.51 |
| VCHM | 48.3 | 2.04 | 47.15 |
| VCHM-Replace | 47.8(↓0.5) | 2.04 | 47.34 |
| OrSM | 48.3 | 2.04 | 47.13 |
| OrSM-Replace | 47.5 (↓0.8) | 2.04 | 47.35 |

Table E. Performance Comparison of Fusion Methods.

| Method | $mAP_{50}$ | Params | GFLOPs |
|---|---|---|---|
| Concat Plus 1x1 conv | 49.7 | 2.66 | 54.53 |
| Cross-attention | 49.8 | 2.52 | 52.44 |
| SE-style cross gating | 50.0 | 2.56 | 52.43 |
| FFM | 50.5 | 2.59 | 52.44 |

Table F. Performance Comparison of Multi-Scale Fusion Methods.

| Method | $mAP_{50}$ | Params | GFLOPs |
|---|---|---|---|
| FPN | 48.2 | 1.59 | 32.28 |
| Enhanced PANet | 50.4 | 2.63 | 52.86 |
| Ours | 50.5 | 2.59 | 52.44 |

**Generalization Capability.** To validate the generalization capability of BDNet, we conducted a cross-dataset evaluation by training on the AI-TODv2 dataset and testing on the VisDrone2019 dataset. The category mapping from VisDrone2019 to AI-TODv2 is as follows: pedestrian/people → person; car/van/truck/bus → vehicle; redundant categories were discarded. The results in Tab. G demonstrate that BDNet exhibits strong generalization capability.

Table G. Cross-Dataset Validation.

| Method | Vehicle | Person | $mAP_{50}$ |
|---|---|---|---|
| YOLO11 | 57.0 | 23.6 | 40.3 |
| YOLO12 | 62.6 | 35.8 | 49.2 |
| Ours | **64.9** | **36.3** | **50.6** |

**Validation of Deployment Value.** Tab. H reports the model's inference speed (FPS) and memory usage across different hardware platforms (e.g., GPU and edge devices). The tests were conducted with an input resolution of 640×640 and a batch size of 1. The results demonstrate that the model achieves high accuracy while maintaining practical deployment value.

Table H. GPU/Edge Devices Deployment Value.

| Model | Peak Memory Usage | FPS | Device |
|---|---|---|---|
| YOLO12-I | 1191.05MB | 18.55 | NVIDIA RTX 3090Ti |
| BDNet (Ours) | 1072.85MB | 21.98 | NVIDIA RTX 3090Ti |
| YOLO12-I | 2949.65MB | 9.19 | Jetson Orin Nano |
| BDNet (Ours) | 2810.02MB | 10.16 | Jetson Orin Nano |

## References


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